

# RESUME

## RESUME — HelixLLM Full-Extension Programme (session-resumption file, §11.4.131)

**Fresh session? Paste the SHORT prompt below.** This file is the single out-of-the-box entry point. Always current; re-read on every new session. Track (T1/main).

### SHORT resumption prompt (paste into a fresh session)

Read docs/research/07.2026/00\_master/RESUME.md + .superpowers/sdd/progress.md (the live SDD ledger), run `git fetch --all`, and continue the HelixLLM full-extension programme. Phase 0 (GPU) + Phase 1 (fleet + fixes) + Phase 2 (HelixAgent → HelixLLM e2e) are PROVEN; the LLMsVerifier capability-verification chain (C1 → C2 → C4 → C5 → C3) is fully landed and under a final combined independent review. Next: land the review to GO, then the release-prep pass (pointer bumps + prefixed release tag §11.4.151 via merge-onto-latest §11.4.113, NO force-push). Honor anti-bluff §11.4, subagent-driven §11.4.70, (T1/<branch>) labels §11.4.182, one canonical branch feature/helixllm-full-extension §11.4.181, §11.4.174 shared-host process/tree ownership.

### Current phase + immediate next action

- **Phase 0 (GPU foundation):** COMPLETE — rootless CDI passthrough + sm\_120 build + real 30B inference PROVEN.
- **Phase 1 (fleet + fixes):** 30B coder live; Containerfile + claude\_toolkit fixes landed, all re-reviewed GO.
- **Phase 2 (HelixAgent → HelixLLM e2e):** PROVEN + review GO — real generate + Postgres/Redis persistence (cognee/vector honest SKIP, OQ2).
- **LLMsVerifier chain:** C1 C2 C4 C5 C3 all landed — **under final combined C4+C5+C3 review** (in flight).
- **Immediate next:** land the combined review to GO (or one fix → re-review loop); then **release-prep** — refresh this file (done), bump meta-repo

submodule pointers (careful of the §11.4.174 concurrent helix\_agent go.mod track), then the prefixed release tag.

- **Terminal goal (this scope):** a fully-validated, prefixed release tag (§11.4.151) published across main + all owned submodules; local HelixLLM on the RTX 5090 exposed via HelixAgent to HelixCode/CLI agents.

## LIVE SERVER (operator-testable, running now)

**Qwen3-Coder-30B-A3B serving on `http://localhost:18434/v1`** (OpenAI-compatible) — container `helixllm-coder` (`podman ps`), `--network=host`, image `localhost/helixllm/llamacpp-router:cuda12.8-sm120`, 8 parallel slots, 24k ctx, q8\_0 KV, ~19.4 GB VRAM. PROVEN: single-stream ~220 tok/s (`coder_live_e2e` log; RESUME's older ~322 tok/s was a different `-jinja/q8_0-KV` run) + 8 concurrent agents @85–96 tok/s; real coding output (`is_palindrome`, `func Add`). Restart: `podman start helixllm-coder`.

**ENDPOINT GOTCHA (load-bearing, proven in Phase-2 docs/qa/phase2\_e2e\_20260706/12\_endpoint\_finding.txt):** raw `curl` uses `http://localhost:18434/v1/chat/completions`. A client/SDK that APPENDS `/v1/chat/completions` (incl. HelixAgent via `HELIX_LLM_LOCAL_OPENAI_ENDPOINT`) MUST use the BASE `http://localhost:18434` (NO `/v1`) — else `.../v1` → double `/v1/v1` → **HTTP 404**. (This SUPERSEDES the `:18434/v1` value in `10_llmsverifier_helixagent/PHASE2_BLOCKERS_INVESTIGATION.md` OQ1 — that doc line is pending a correction commit in the release-gate sweep.) NOTE: latest `llama.cpp` `-fa` takes a value (`-fa on`).

## Live-state anchors (facts, §11.4.6)

| Anchor                       | Value                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------|
| Constitution HEAD followed   | 0882b9e (through §11.4.182)                                                                                |
| Host                         | ALT Workstation 11.1; RTX 5090 32 GB; driver 570.169; CUDA 12.8; podman 5.7.1 rootless; 64 cores / 251 GiB |
| Canonical branch (§11.4.181) | feature/helixllm-full-extension — ACTIVE (no upstream tracking configured yet)                             |
| Router image (built+proven)  | localhost/helixllm/llamacpp-router:cuda12.8-sm120 — latest llama.cpp, sm_120, OpenSSL/curl                 |

| Anchor                          | Value                                                                                                                                 |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
|                                 | (-hf HTTPS proven), ships rpc-server                                                                                                  |
| Release prefix (§11.4.151)      | HELIX_RELEASE_PREFIX else helix_code                                                                                                  |
| Shared-host caution (§11.4.174) | helix_agent checkout carries a CONCURRENT QA/dep track's uncommitted go.mod/go.sum + .qa_bak — NOT ours; do not sweep on pointer-bump |

## Done so far this session (real, evidence-backed commits)

- helix\_llm 13d2d27 Containerfile: hard-fail ggml-rpc-server copy (§11.4.122) + OpenSSL/curl — re-review **GO**; d8b3fa2 phase-1 QA evidence (-hf 469MB HTTPS download + live coder e2e); 3f85e3d5 OPERATOR\_GUIDE.
- claude\_toolkit ef77b19 loud-fail resolution + discriminating test; 9d12347 C1 regression guard (37/0, §1.1 mutation-proven) — **REVIEW-3 CLEAN**.
- llms\_verifier 09f9533c C4 (RAG/Skills/Plugins probes) · 28e6625a C5 (Verify composes+persists VerificationResult) · ad18e91f C3 (fail-closed resolver, seeds → unverified) — **chain complete, under combined review**.
- helix\_agent 17f08ba9 Phase-2 live tests (provider e2e + redis) — pointer NOT yet bumped (§11.4.174).
- helix\_code cf26b813 embeddings design (TEI CPU) · 9bf4c3da Phase-2 stack design (Qdrant) · da0fabae blockers investigation (OQ1/OQ2) · 5223d10d C3 handoff · 278df582 Phase-2 e2e evidence.

## Next actions (in order)

1. Land the combined **C4+C5+C3 review** to GO (adjudicate detector.go:61 no-probe self-cert).
2. Correct PHASE2\_BLOCKERS\_INVESTIGATION.md OQ1 verdict → base <http://localhost:18434>.
3. Release-prep: bump main-repo submodule pointers (helix\_llm, claude\_toolkit, llms\_verifier; helix\_agent carefully per §11.4.174) → §11.4.40 full-suite pre-tag sweep → prefixed tag (§11.4.151) → publish via merge-onto-latest (§11.4.113, NO force-push §2.1 multi-upstream).
4. Post-release follow-ups (tracked): P-OQ2-A wire cognee\_memory\_repository + P-OQ2-B re-verify cognee 1.2.2 bug;

detector.go:61 no-probe self-cert; VRAM broker IMPL (design 5102607 done) to unlock GPU vision/image/video; Phase-3 CPU embeddings + translation NMT impl (designs done).

## **Binding constraints (do not violate)**

Anti-bluff §11.4 (real captured proof, no metadata PASS) · runtime-signature = done §11.4.108 · no-force-push merge-onto-latest §11.4.113 · rootless podman via containers submodule §11.4.76/§11.4.161 · no CI §11.4.156 · no silent removal §11.4.122 / investigate-before-remove §11.4.124 · deep multi-angle research per change §11.4.150 · independent review to GO §11.4.125/§11.4.142/§11.4.134 · one canonical branch §11.4.181 · (T<N>/<branch>) labels §11.4.182 · shared-host ownership §11.4.174.